

# HVAC parts list - 340 Ridgeway Pl, Sedro-Woolley WA 98284 (Glenhaven Lakes, unincorporated Whatcom County)

SFR2025-00214 model **340-ridgeway-hvac** printed 2026-09-25 ductless heat 25,039 Btuh cooling 1.64 t 1 outdoor 5 heads 0 ducted units 0 registers line sets 253 ft duct 0 ft vent 47 ft 9 days on site  
Design 19 F / 78 F (sedro-woolley: WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Skagit foothills: the Bellingham row is the colder neighbour - STAND-IN)). Foundation crawl, line sets concealed, ducts inside, ventilation exhaust. Quantities only - the prices are set on the price list in the app.

## Loads

heating 25,039 Btuh (10.3 Btuh/sf) · cooling 18,820 + 823 latent · WSU sheet 31,752 Btuh · infiltration 2 ACH50 -> 0.179 ACH at design · ventilation 70.3 cfm (exhaust)

| room                    | storey | sf  | heat Btuh | cool Btuh | glass sf | ext wall ft |
|-------------------------|--------|-----|-----------|-----------|----------|-------------|
| MASTER                  | upper  | 279 | 3,842     | 3,577     | 140      | 53          |
| OFFICE                  | main   | 108 | 3,506     | 4,672     | 185      | 42          |
| MEDIA                   | upper  | 266 | 3,256     | 2,809     | 131      | 49          |
| ENTRY                   | main   | 257 | 2,452     | 1,511     | 86       | 36          |
| BED. 2                  | upper  | 167 | 1,668     | 958       | 36       | 30          |
| KITCHEN                 | main   | 122 | 1,382     | 2,202     | 31       | 25          |
| DINING                  | main   | 193 | 1,101     | 570       | 15       | 12          |
| LANDING / OPEN TO BELOW | upper  | 163 | 971       | 363       | 20       | 16          |
| HALL                    | main   | 50  | 946       | 262       | 42       | 13          |
| WINDOW SEAT             | main   | 20  | 721       | 745       | 18       | 15          |
| GARAGE STORAGE          | main   | 29  | 619       | 274       | 15       | 12          |
| SHOWER                  | upper  | 20  | 429       | 297       | 12       | 9           |
| LAUNDRY                 | main   | 49  | 367       | 37        | 2        | 8           |
| MASTER                  | upper  | 68  | 365       | 122       | 7        | 6           |
| SHELF / NICHE           | upper  | 12  | 245       | 16        | 0        | 7           |
| CLOS                    | main   | 13  | 190       | 3         | 0        | 5           |
| MASTER                  | upper  | 20  | 140       | 69        | 3        | 2           |
| COAT                    | main   | 22  | 94        | 1         | 0        | 2           |
| HALL                    | upper  | 51  | 87        | 56        | 0        | 0           |
| W.I.C                   | upper  | 36  | 53        | 39        | 0        | 0           |
| MASTER BATH             | upper  | 12  | 44        | 22        | 0        | 1           |
| BATH. 2                 | upper  | 17  | 29        | 18        | 0        | 0           |
| PWDR                    | main   | 15  | 21        | 0         | 0        | 0           |
| LINEN / STOR.           | upper  | 10  | 15        | 11        | 0        | 0           |
| PANTRY                  | main   | 10  | 14        | 0         | 0        | 0           |

## The system

**unit house - ductless:** heat 25,039 Btuh, capacity at design 33,595 Btuh  
outdoor eq\_mz\_36 36k (multi) on the W wall, at the electrician's Ductless mini-split outdoor unit (DINING, electrical/jobs/ridgeway/devices.json); 253 ft of line set

| head in | zone rooms             | carried                                      | heat  | cool  |
|---------|------------------------|----------------------------------------------|-------|-------|
| DINING  | KITCHEN, DINING, ENTRY | CLOS, PANTRY, WINDOW SEAT, HALL, LAUNDRY     | 7,962 | 5,330 |
| OFFICE  | OFFICE                 | PWDR, GARAGE STORAGE, COAT                   | 4,707 | 4,947 |
| MASTER  | MASTER                 | SHELF / NICHE, MASTER BATH, SHOWER           | 5,622 | 4,103 |
| MEDIA   | MEDIA                  | LINEN / STOR., LANDING / OPEN TO BELOW, HALL | 4,805 | 3,239 |
| BED. 2  | BED. 2                 | W.I.C, BATH. 2                               | 1,943 | 1,015 |

| head | room   | storey | heat  | cool  |
|------|--------|--------|-------|-------|
| 9k   | DINING | main   | 7,962 | 5,330 |
| 6k   | OFFICE | main   | 4,707 | 4,947 |
| 9k   | MASTER | upper  | 5,622 | 4,103 |
| 6k   | MEDIA  | upper  | 4,805 | 3,239 |
| 6k   | BED. 2 | upper  | 1,943 | 1,015 |

## Runs (measured, before waste)

| system / size | LF |
|---------------|----|
|---------------|----|

|                        |     |
|------------------------|-----|
| condensate 0.75        | 101 |
| dryer 4                | 13  |
| exhaust 4              | 17  |
| exhaust 6              | 13  |
| hood 8                 | 4   |
| refrigerant 0.25x0.375 | 253 |

## Equipment

### Equipment

| item                                                              | measured | order | unit | note                                                                                                           |
|-------------------------------------------------------------------|----------|-------|------|----------------------------------------------------------------------------------------------------------------|
| Ductless wall head 6k                                             | 3        | 3     | EA   | OFFICE: zone OFFICE - heat 4,707 / cool 4,947 Btuh · MEDIA: zone MEDIA - heat 4,805 / cool 3,239 Btuh · BED. 2 |
| Ductless wall head 9k                                             | 2        | 2     | EA   | DINING: zone KITCHEN, DINING, ENTRY - heat 7,962 / cool 5,330 Btuh · MASTER: zone MASTER - heat 5,622 / cool 4 |
| Multi-zone outdoor unit - 36k, cold-climate, up to 5 indoor units | 1        | 1     | EA   | multi-zone outdoor unit, 36k nominal, ~33,595 Btuh at 19 F (ResStock cold-climate curve), at the electrician's |
| Wired wall controller per ductless zone (optional) (option)       | 5        | 5     | EA   | wired wall controller per head (option)                                                                        |

## Rough-in

### Condensate

| item                                                        | measured | order | unit | note                                                                             |
|-------------------------------------------------------------|----------|-------|------|----------------------------------------------------------------------------------|
| Condensate line 3/4 in PVC (or 5/8 vinyl with the line set) | 101.2    | 112   | LF   | condensate: gravity, 1/8 in per ft, to daylight (M1411.3) - never into the crawl |

### Refrigerant piping

| item                                                                 | measured | order | unit | note                                                                                                           |
|----------------------------------------------------------------------|----------|-------|------|----------------------------------------------------------------------------------------------------------------|
| Anti-vibration pads (set of 4)                                       | 1        | 1     | EA   |                                                                                                                |
| Communication cable 14/4 stranded (outdoor unit to each indoor unit) | 253.5    | 279   | LF   | with the line set (powers and talks to the head)                                                               |
| Equipment pad - composite, sized to the outdoor unit                 | 1        | 1     | EA   | outdoor unit pad                                                                                               |
| Line set 1/4 x 3/8 insulated (pre-flared or brazed)                  | 253.5    | 279   | LF   | DINING 9k head, 19 ft concealed · OFFICE 6k head, 59 ft concealed · MASTER 9k head, 86 ft concealed · MEDIA 6k |
| Outdoor unit stand / wall bracket (snow and drainage clearance)      | 1        | 1     | EA   | snow / drainage stand (NW practice: 12-18 in clear of grade)                                                   |
| Wall penetration - 3 in sleeve, sealed and flashed                   | 1        | 1     | EA   | rim-joint penetration for the line-set bundle beside the outdoor unit                                          |

### Support + protection

| item                                                              | measured | order | unit | note                                                            |
|-------------------------------------------------------------------|----------|-------|------|-----------------------------------------------------------------|
| Bore through a stud / plate (counted, not priced)                 | 16       | 16    | EA   | plates the line set passes (bottom plate + subfloor per storey) |
| Fire caulk / sealant (tube)                                       | 2        | 2     | EA   | fire caulk / sealant at the penetrations                        |
| Head / controller blocking (framer or HVAC)                       | 5        | 5     | EA   | head mounting plate blocking                                    |
| Nail plate at a line-set / cable bore within 1-1/4 in of the face | 16       | 16    | EA   | nail plates at those bores                                      |

## Ventilation and exhaust

### Ventilation

| item                                                                                                                             | measured | order | unit | note                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------|----------|-------|------|----------------------------------------------------------------------------------------------------------------|
| Bath fan exhaust duct 4 in insulated flex (50 cfm fan: <= 25 ft, Table M1505.4.4.2)                                              | 16.8     | 19    | LF   | PWDR: 50 cfm, 7 ft, 2 elbows -> 4 in flex (Table M1505.4.4.2 allows 25 ft) · MASTER BATH: 50 cfm, 6 ft, 3 elbo |
| Dryer duct 4 in 90 elbow                                                                                                         | 2        | 2     | EA   |                                                                                                                |
| Dryer duct 4 in rigid metal, smooth wall (M1502.4.2)                                                                             | 13.1     | 15    | LF   | from the electrician's dryer outlet in LAUNDRY: 13 ft                                                          |
| Dryer vent box (recessed)                                                                                                        | 1        | 1     | EA   |                                                                                                                |
| Dryer wall cap with damper, no screen (M1502.3)                                                                                  | 1        | 1     | EA   |                                                                                                                |
| Exhaust duct 6 in insulated (80-110 cfm fans, whole-house fan: Table M1505.4.4.2)                                                | 13.2     | 15    | LF   | BATH. 2 whole-house fan: 90 cfm, 13 ft, 3 elbows -> 6 in flex (Table M1505.4.4.2 allows 45 ft)                 |
| Make-up air damper (motorised) + hood interlock + duct (WA M1503.6: hood > 400 cfm with a non-direct-vent fuel appliance inside) | 1        | 1     | EA   | 600 cfm hood > 400 cfm: motorised make-up air damper interlocked with the hood - the plan calls for it and WA  |

|                                                                                                                           |      |    |    |                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------|------|----|----|-----------------------------------------------------------------------------------------------------------------|
| Outdoor air inlet 4 sq in NFA, through-wall with screen (WA exhaust-only whole-house ventilation, one per habitable room) | 5    | 5  | EA | unit house: an outdoor-air inlet (4 sq in NFA) in each of 5 habitable spaces (WA exhaust-only systems) - window |
| Range hood duct 6-10 in rigid, sealed (M1503.3)                                                                           | 16.2 | 18 | LF | 8 in rigid for a 600 cfm hood at the electrician's range outlet (electrical/jobs/ridgeway/devices.json) (M1503) |
| Range hood wall / roof cap with damper                                                                                    | 1    | 1  | EA | 8 in wall cap with damper                                                                                       |
| Wall / roof cap 6 in with backdraft damper                                                                                | 1    | 1  | EA | BATH. 2 whole-house fan: wall cap with backdraft damper                                                         |
| Wall cap 4 in with backdraft damper                                                                                       | 3    | 3  | EA | PWDR: wall cap with backdraft damper · MASTER BATH: wall cap with backdraft damper · SHOWER: wall cap with bac  |
| Whole-house exhaust fan, continuous duty, <= 1.0 sone (M1505.4) - replaces one electrical allowance fan                   | 1    | 1  | EA | unit house: whole-house fan 70 cfm continuous (Qr 47 x Csystem 1.5, one fan = unbalanced, not distributed) in   |
| Whole-house ventilation control - 24 h timer / override, labelled (WA M1505.4.2)                                          | 1    | 1  | EA | labelled "WHOLE HOUSE VENTILATION - LEAVE ON" 24 h control with manual override (WA M1505.4.2); the electrician |

## Start-up, tests and design

### Commissioning

| item                                                                                        | measured | order | unit | note                                                                                                   |
|---------------------------------------------------------------------------------------------|----------|-------|------|--------------------------------------------------------------------------------------------------------|
| Equipment labels, WSEC certificate data, owner hand-over                                    | 1        | 1     | EA   | per dwelling unit                                                                                      |
| Load calculation + equipment selection for the permit (Manual J / S or the WSU sizing form) | 1        | 1     | EA   | room-by-room loads for the permit (this takeoff is the draft)                                          |
| Start-up, charge verification, controls set-up (per outdoor unit)                           | 1        | 1     | EA   | 36k system                                                                                             |
| Ventilation flow test - whole-house + local exhaust (WA M1505.4.3 / M1505.4.4)              | 1        | 1     | EA   | unit house: whole-house 70 cfm + every local exhaust measured at the grille (WA M1505.4.3 / M1505.4.4) |

### Refrigerant piping

| item                                                           | measured | order | unit | note                                                       |
|----------------------------------------------------------------|----------|-------|------|------------------------------------------------------------|
| Nitrogen pressure test + evacuation to 500 micron (per system) | 1        | 1     | EA   | per refrigerant system                                     |
| Refrigerant - additional charge beyond the pre-charge (lb)     | 1.9      | 1.9   | lb   | 30 oz past the pre-charged length [U: the maker's oz / ft] |

## The checker

**PRACTICE** 36k outdoor unit (unit house): it sits under the upper storey (BED. 2 above): out of the snow, but set it on a ground stand with isolation pads, never a wall bracket - compressor vibration carries into the room above

**PRACTICE** MASTER: line set 86 ft > the ~82 ft this unit class allows [U - the model's manual governs] - move the outdoor unit or the head

**PRACTICE** multi-zone: total piping 253 ft > ~229 ft for a 36k multi-zone [U] - split the zones over two outdoor units

**DATA** plan set: ach50: the set prints {1.5: 1, 2.0: 1} -> 1.5 (two values for one requirement; the tighter one is used for ACH50, the most frequent for R-values - confirm on the energy sheet)

**DATA** plan set: credit 2.1 sets 2.0 ACH50 but the notes print 1.5 -> 1.5 (the tighter target governs the blower-door test and the infiltration load)

**DATA** design temperature: 19 F is a stand-in (WSEC Table RC-1 Bellingham IAP 19 F / 78 F [S]; MJ8 Table 1A 23 F / 76 F [V] (Skagit foothills: the Bellingham row is the colder neighbour - STAND-IN)) - pull WSEC Table RC-1 for sedro-woolley before this goes out

**PRACTICE** electrical: the 36k multi outdoor unit typically needs a 30 A circuit [U: the nameplate MOCP governs]; the electrical job carries 20 A

note: tracer fragments merged into one space (same name, touching): MASTER (upper: 3 pieces, 368 sf)

note: under 70 sf (IRC R304.1 - not a habitable room), so carried by a neighbour and never given a head or an ERV supply: WINDOW SEAT 20 sf

note: unit house: 4 bath / powder room(s) have no head - they are carried by the zone they open into; WSEC allows resistance heat up to 0.5 W/sf or 500 W if a bath needs its own (an option line, not in the base)

note: bath fans: 4 read from the electrical skill (electrical/jobs/ridgeway/devices.json) - the electrician supplies and wires them; HVAC ducts and caps each one

note: unit house: the plan's 90 cfm whole-house fan covers the 70 cfm WA rate (Qr 47 x 1.5 for one exhaust fan)

note: equipment must meet HSPF2 >= 9 (the WSEC credit); design temperature 19 F <= 23 F: a NEEP-listed cold-climate variable-capacity unit (the 3.6 note)

note: refrigerant: a split system installed from 1/1/2026 is an A2L (R-454B / R-32) unit under the EPA Technology Transitions rule; R-410A equipment is out. A2L units carry leak detection / mitigation per their listing - it rides with the equipment line

note: a gas-fuel fireplace is on the set - not in this bid (the fireplace installer's); a sealed direct-vent unit needs no combustion air from the house

note: ductless: each head's programmable wireless controller meets WSEC R403.1.1; a wired wall controller per zone is an option line

note: the electrical job carries a separate circuit for the indoor heads (15 A): on Mitsubishi / Fujitsu / Daikin splits the heads are powered from the outdoor unit through the 14/4 - that circuit comes off unless the chosen model needs it

note: electrical: one of its bath-fan allowances becomes the HVAC whole-house fan (supplied here) - strike it from the electrical allowance, keep the wiring and add the 24 h control

## Labour and inspections

crew of 2: set 16.0 h / 3 d, rough 28.0 h / 5 d, startup 5.0 h / 1 d - 9 days at 6.25 h. crew minutes are PLACEHOLDERS (no task-level

install hours were found: references/research-hvac-market.md 3.3) - set them from a sub or the crew

**Mechanical rough-in (cover)** - after framing, before insulation; line sets pressure-tested and in place; ducts, boots, exhaust ducts and caps; firestops at the garage separation

**Ventilation flow test** - before final; whole-house rate and every local exhaust measured (WA M1505.4)

**Final mechanical** - equipment set, started, labelled; equipment listing / HSPF2 on the certificate, controls, condensate, clearances

permit: Whatcom County PDS residential mechanical fee schedule - NOT FOUND in the 9/25 research (references/research-hvac-code.md 8): price it from the county sheet